

Regression Part B

Ulink MM Club
Alan, Frank

Recall: Linear Regression

General formula

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where β_0 denotes the intercept, β_1 denotes the gradient, and ϵ_i denotes the error term (or residue).

Key Assumptions

- ◆ Observations are independent of each other.
- ◆ The residue is normally distributed.

Introduction

Now you are a bank data analyst. With millions of transactions occurring simultaneously within seconds, you are asked to develop an automated system to detect potential fraudulent trading.

How will you construct such a model?

What methods will you use? Is linear regression feasible?

Introduction

In fact, we aim to predict whether a transaction is fraudulent or not, which is a **binary variable** (possible outcomes are 0 and 1). In other words, we are predicting the **probability** of a transaction being fraudulent.

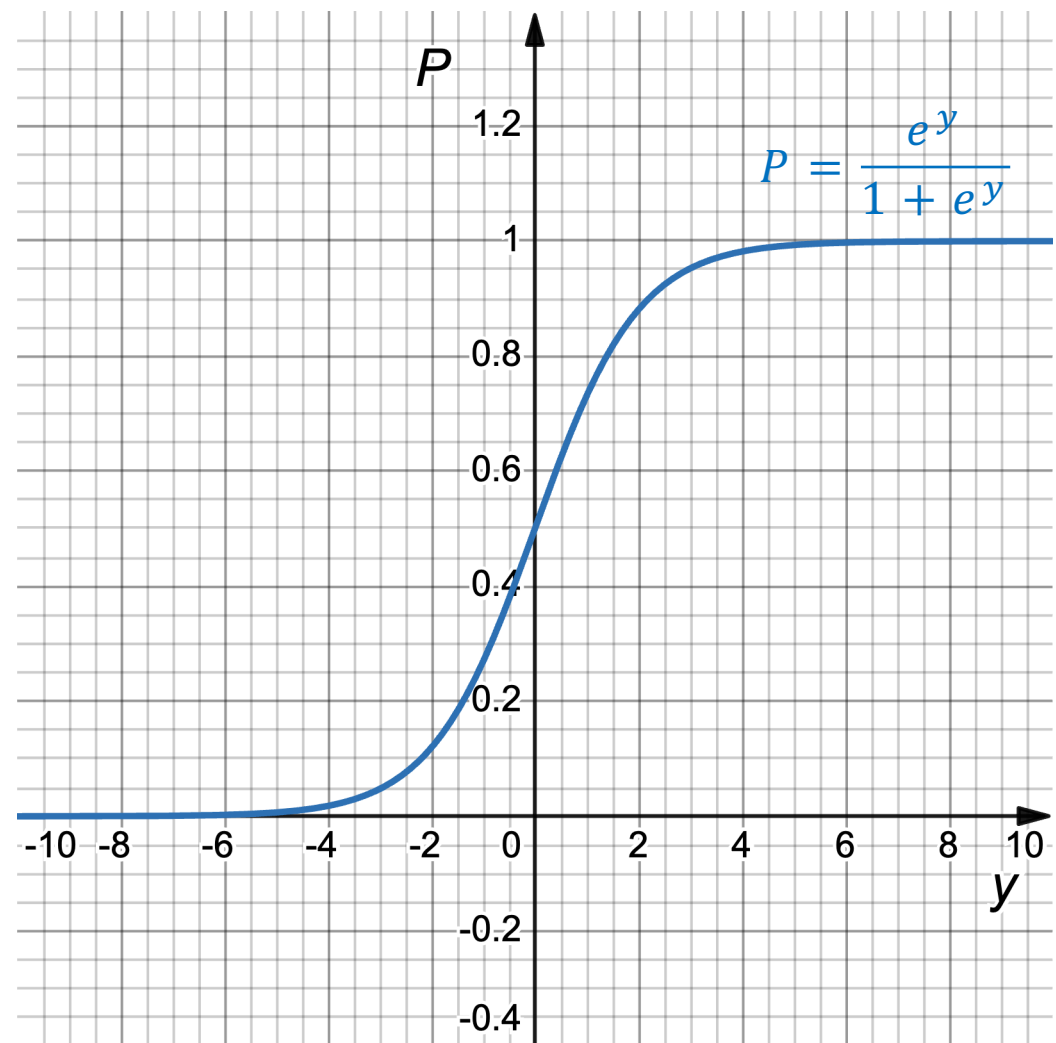
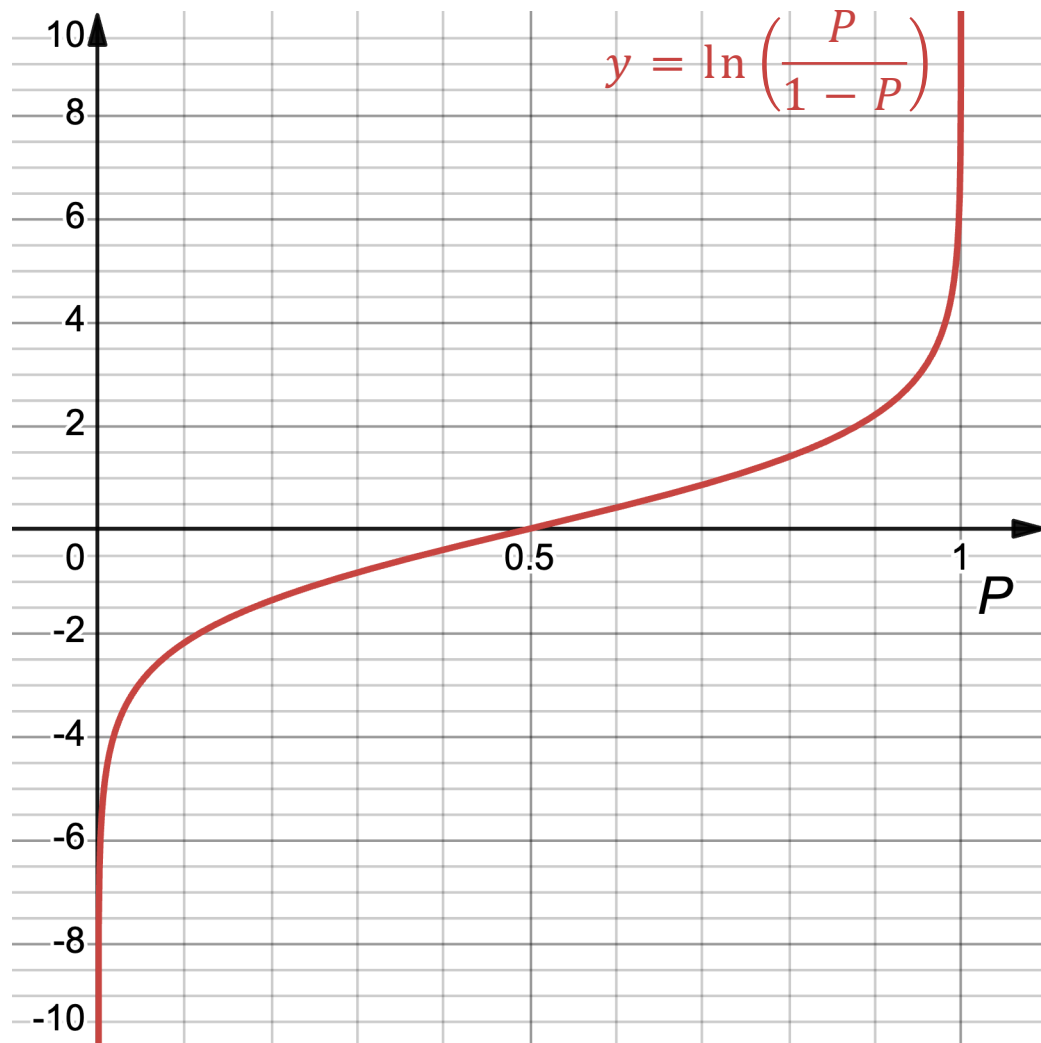
The Logit Transformation

Clearly, our target is in range $[0,1]$, while linear regression targets on continuous dependent variables (which has range $[-\infty, \infty]$). Therefore, we apply a so-called logit transformation:

$$\ln \left(\frac{P}{1 - P} \right)$$

where P denotes the probability.

The Logit Transformation



Logistic Regression

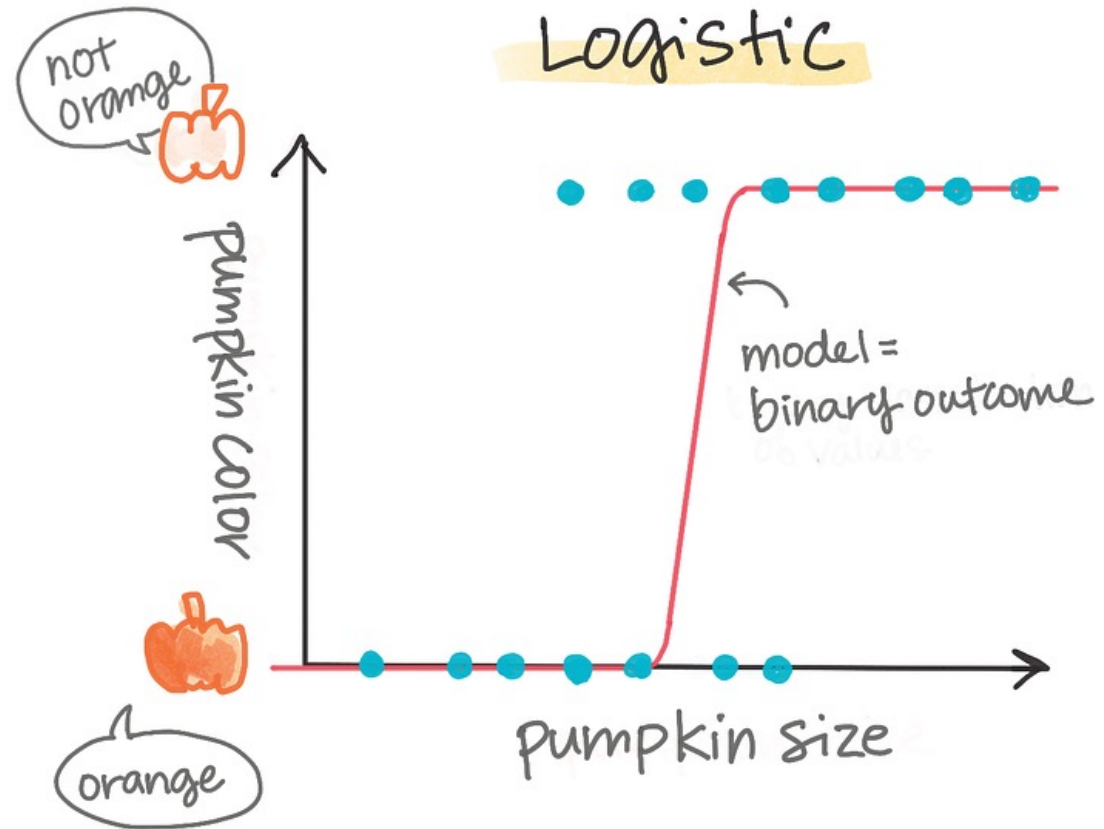
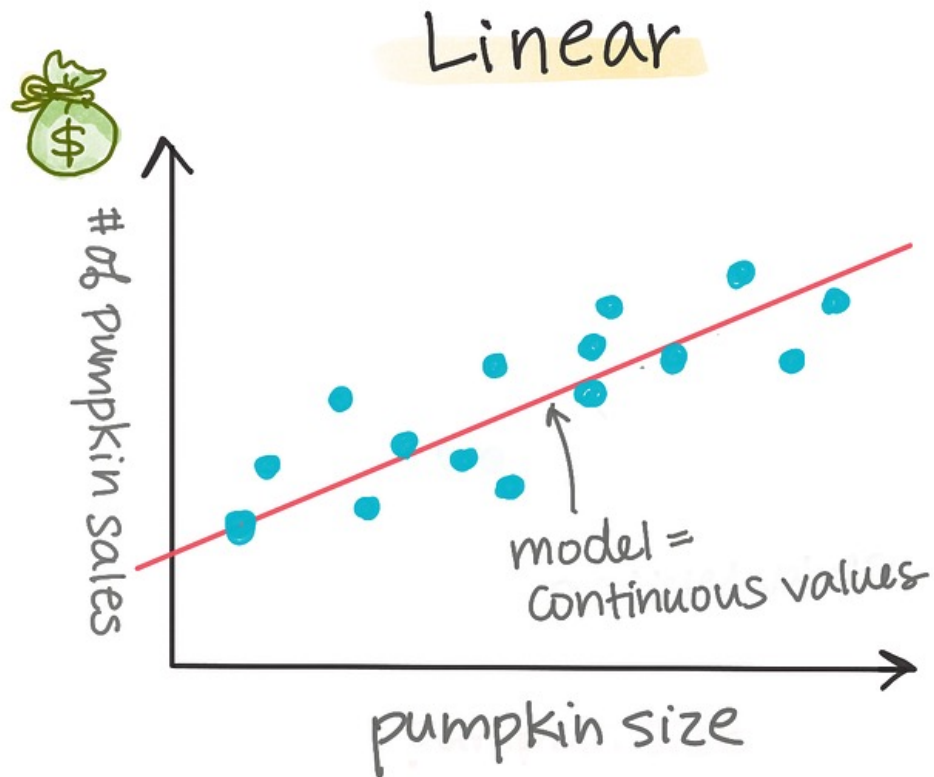
The general formula for logistic regression is given by:

$$\ln \left(\frac{P}{1-P} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

where β_0 denotes the intercept, β_i ($i \geq 1$) denotes the coefficients.

Logistic Regression

LINEAR vs. LOGISTIC REGRESSION



Coefficient Interpretation

Given $\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$, we have odds $\left(\frac{P}{1-P}\right)$ being $e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n}$. Hence, a unit increase in X_i will increase (decrease) the odds to e^{β_i} times if β_i is greater (smaller) than 0.

e.g. studying the effect of age (X_1) on a certain disease (probability as P), we have $\beta_1 = 0.2$. Then 1-year elder increases the odds of disease to $e^{0.2} \approx 1.22$ times, which corresponds to an increase of 22%.

Maximum Likelihood Estimation

For logistic regression, there's no closed formed solution to the maximum likelihood estimation. Therefore, iterative calculations must be conducted by computer programmes.

$$\mathcal{L}(\beta) = \sum_{i=1}^n [y_i \ln(P_i) + (1 - y_i) \ln(1 - P_i)]$$

Evaluating the Models

Akaike Information Criterion (AIC)

$$AIC = 2K - 2\mathcal{L}$$

where K denotes the number of predictors and $\mathcal{L}$ denotes the log likelihood.

Models with smaller AIC values are considered better, note that AIC can only be applied to identify the “optimal” model among several different ones.

There's no absolute criteria for this evaluation indicator.

Evaluating the Models

Hosmer-Lemeshow Test (Goodness of Fit Test)

H_0 : the model is a good fit, H_1 : the model is not a good fit.

Test statistic $H = \sum_{g=1}^G \frac{(O_{1g} - E_{1g})^2}{N_g \pi_g (1 - \pi_g)}$, which follows a χ^2 distribution with degree of freedom $G - 2$. We do not reject H_0 if p value is greater than 0.05 and hence the model is a good fit.

Strengths and Weaknesses

Strengths:

- ◆ The model gives a well-calibrated probability value instead of a simple Y/N;
- ◆ Coefficient interpretability, in terms of odds;

Weaknesses:

- ◆ Sensitive to outliers and correlated predictors;
- ◆ Assumption of linearity between log-odds and predictors, require feature engineering beforehand.

The background is a dark blue, almost black, space filled with a complex, three-dimensional grid of lines that recede into the distance, creating a sense of depth. Overlaid on this grid are numerous mathematical formulas and symbols in a light blue or white color. Some of the visible formulas include the binomial theorem $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$, the Taylor series for $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, the definition of the gamma function $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$, and various summation and integral notations. The formulas are scattered across the image, some appearing larger and more prominent than others, contributing to a dense, intellectual atmosphere.

Thank you